Elective Course

Course Code: CS4103

Autumn 2026-27

Lecture #16

# Artificial Intelligence for Data Science

**Module-2:** PROBLEM SOLVING BY SEARCH

Adversarial Search Problem--- Games [Practice Example]

**Module-3:** CONSTRAINT SATISFACTION PROBLEM (CSP)

Introduction


Course Instructor:

**Dr. Monidipa Das**

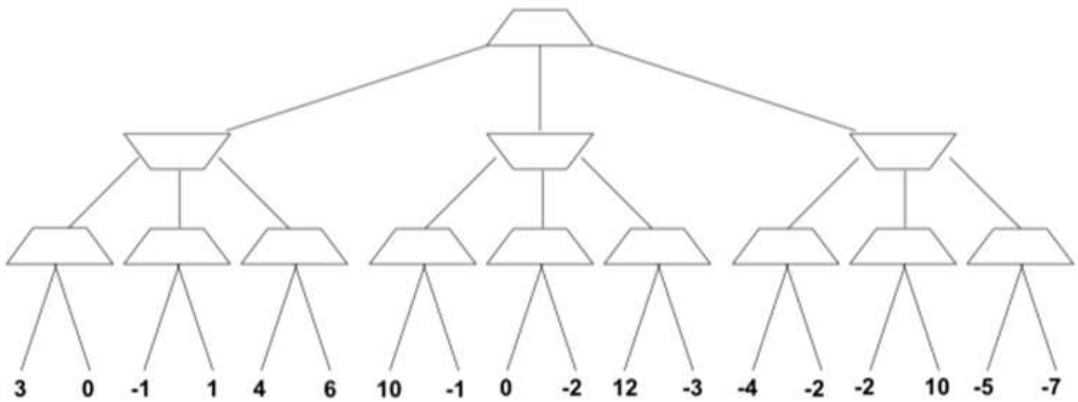
**Assistant Professor**

**Department of Computational and Data Sciences**

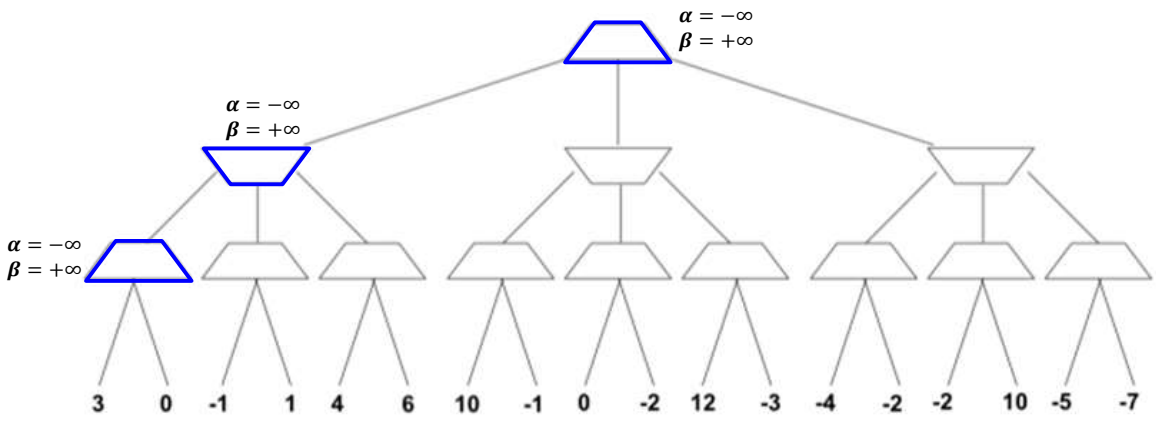
**Indian Institute of Science Education and Research Kolkata, India 741246**


## Practice Problem

**Which leaf nodes are never visited due to pruning?**

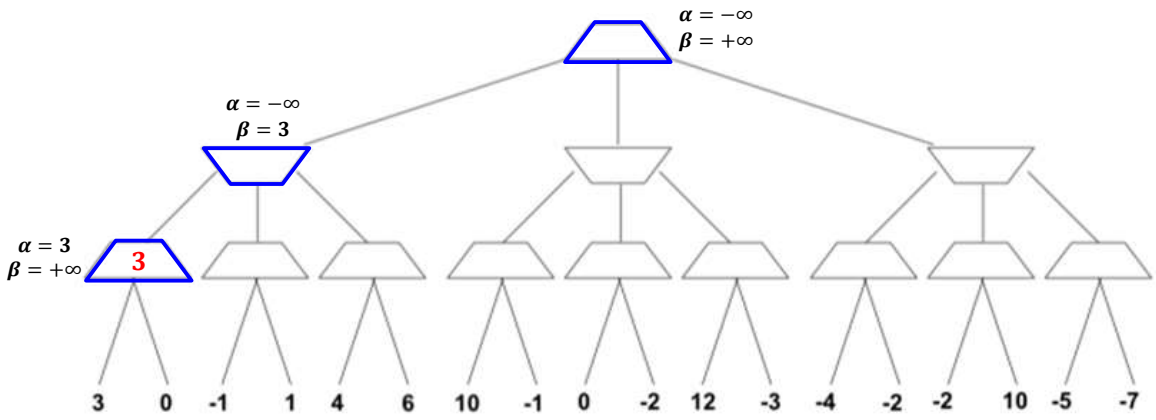


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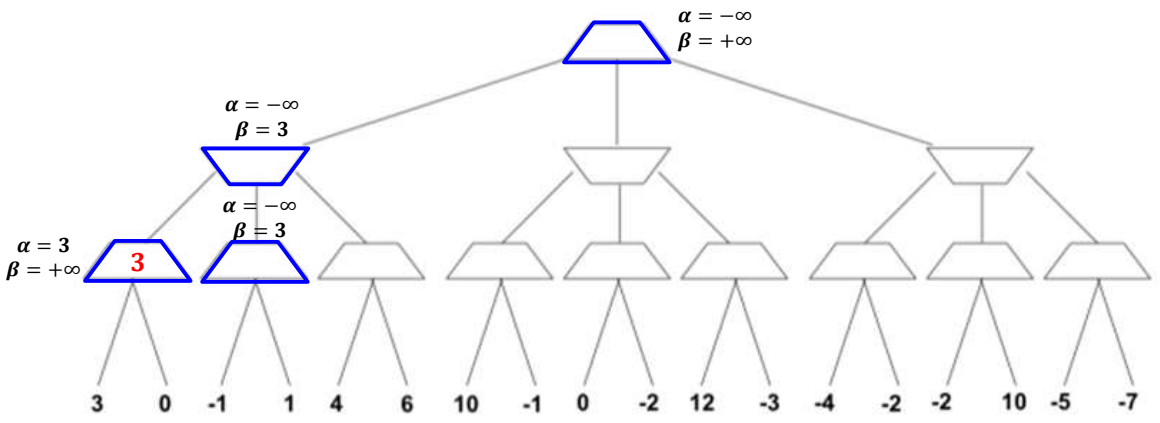





# Solution

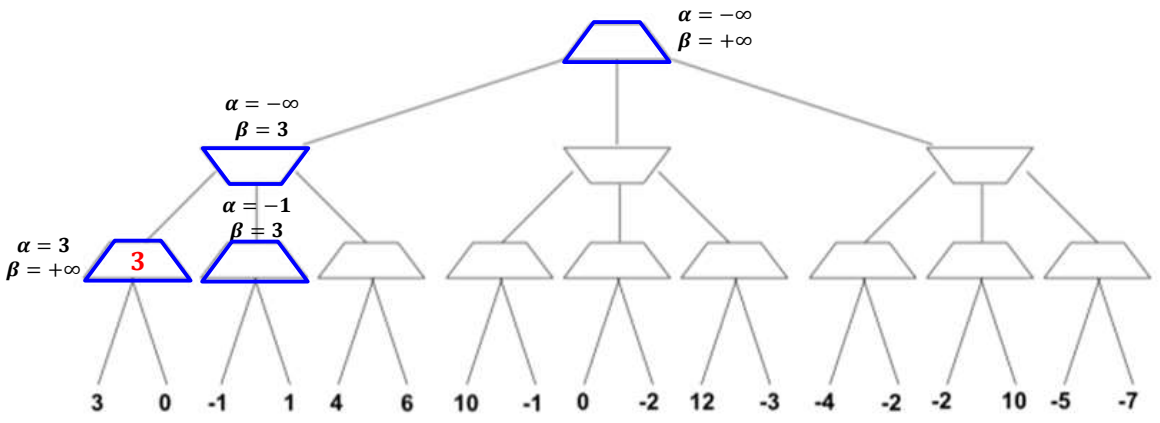


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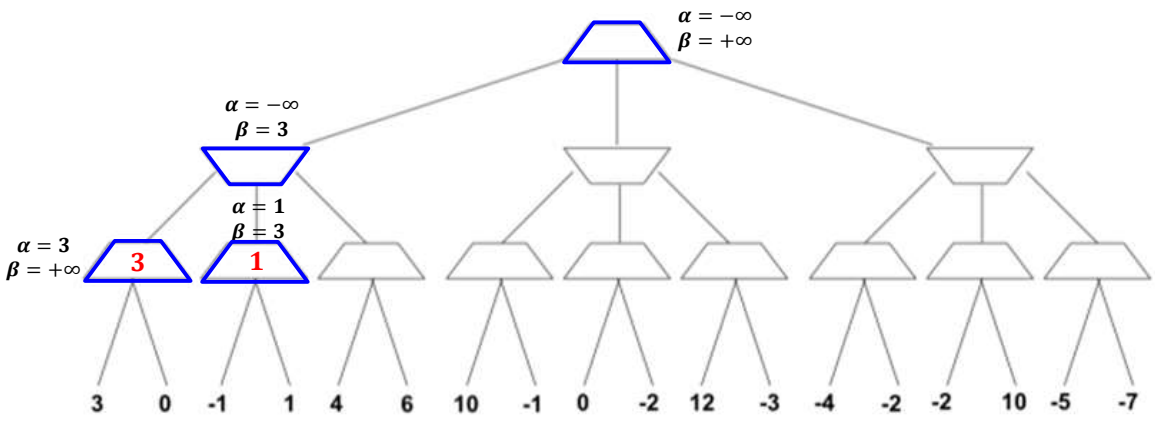





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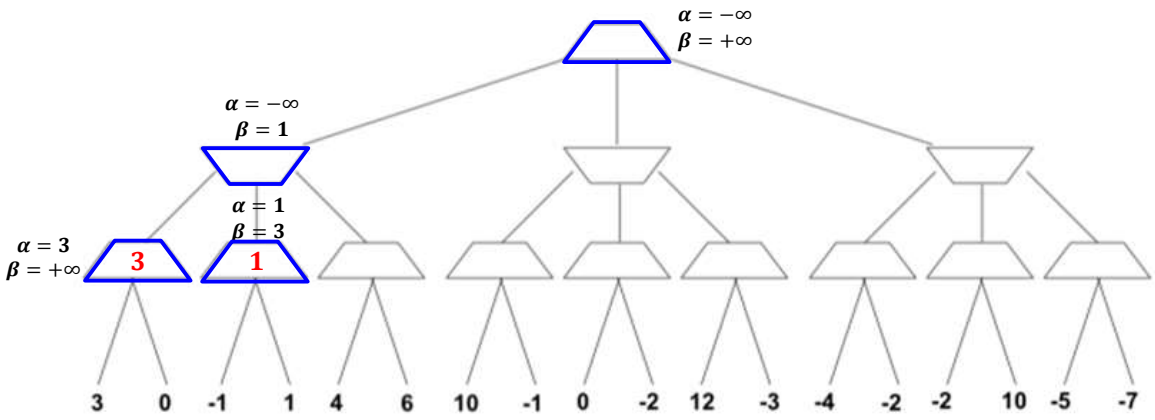


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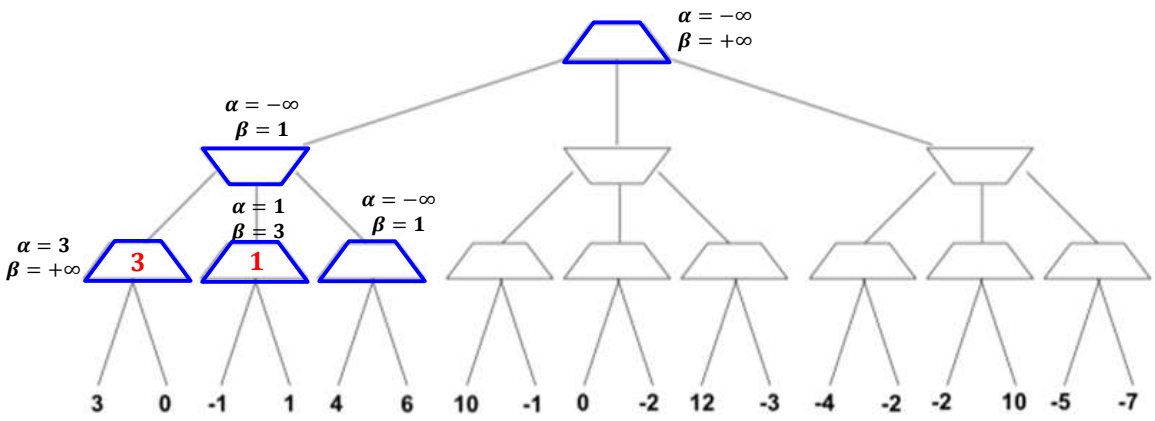



# Solution



# Solution

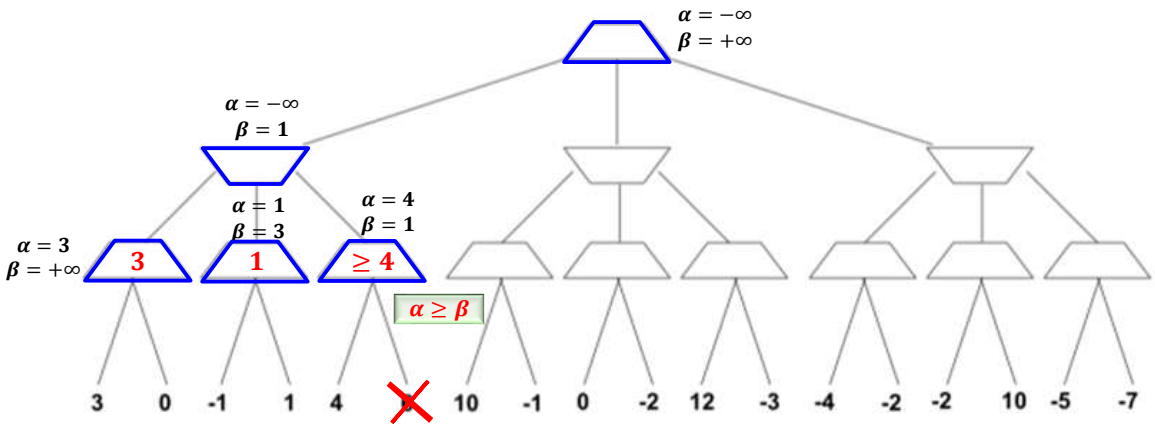
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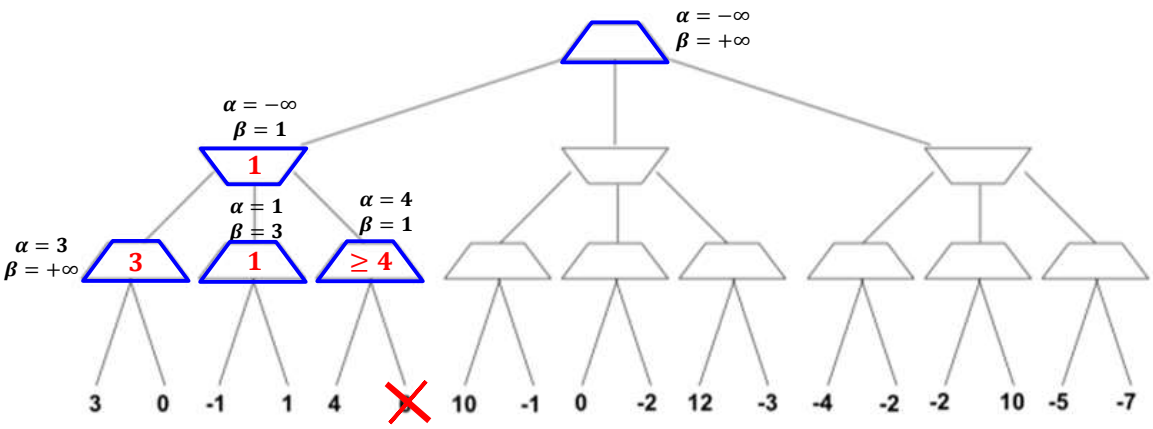



# Solution

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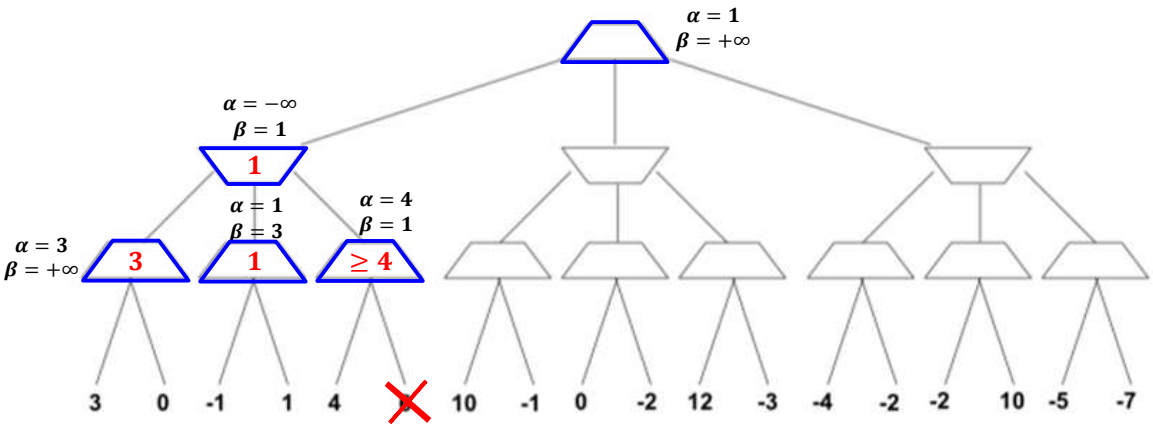


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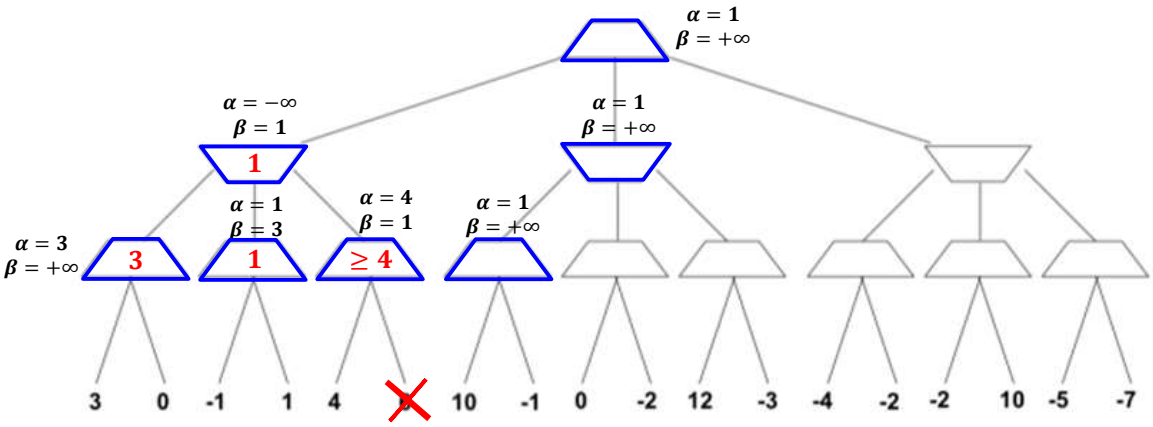





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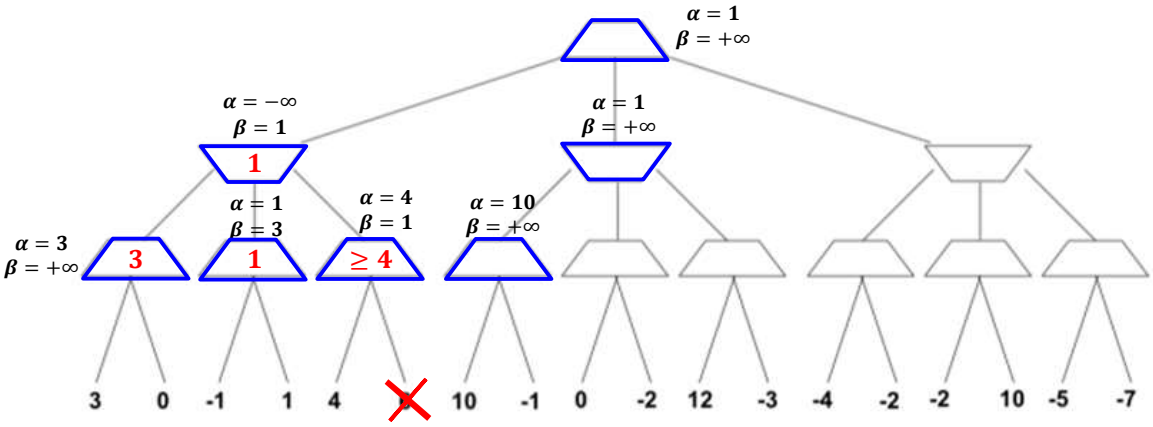


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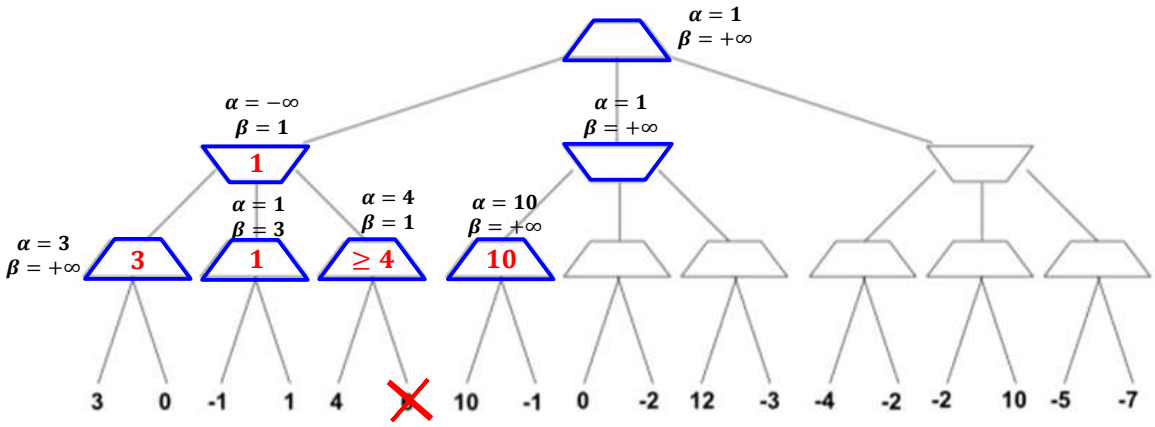





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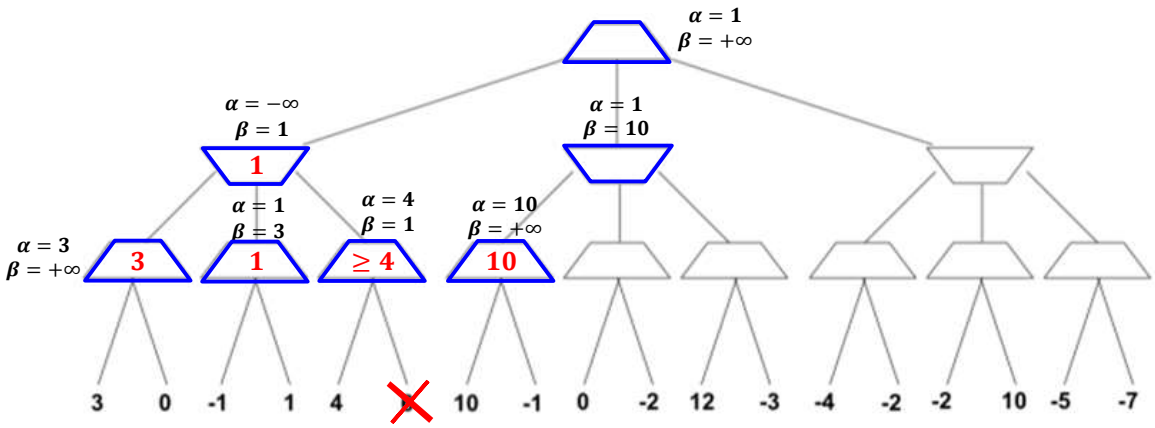


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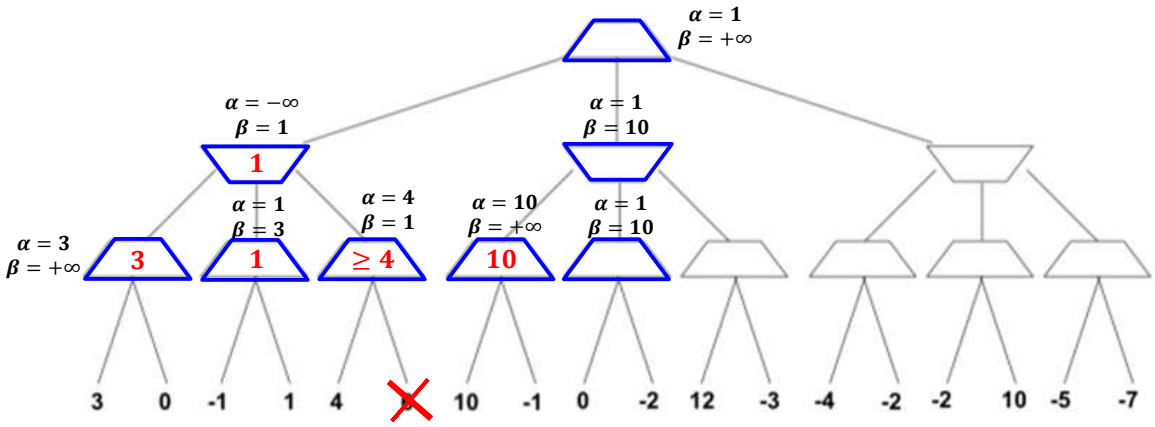





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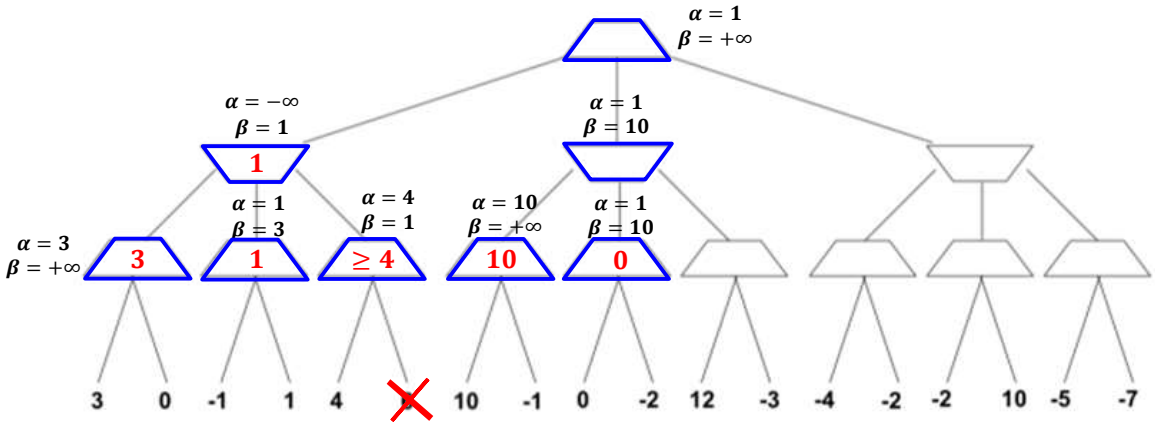


# Solution



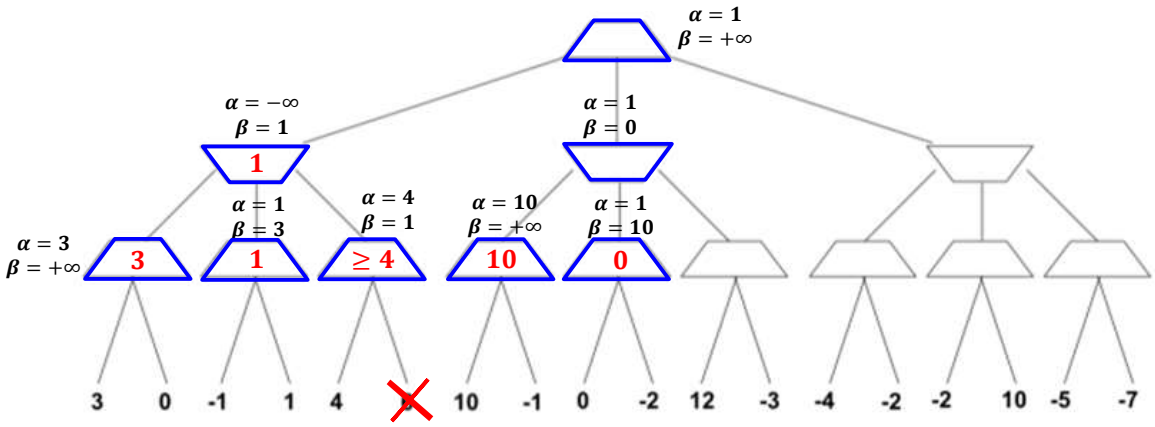



# Solution



# Solution

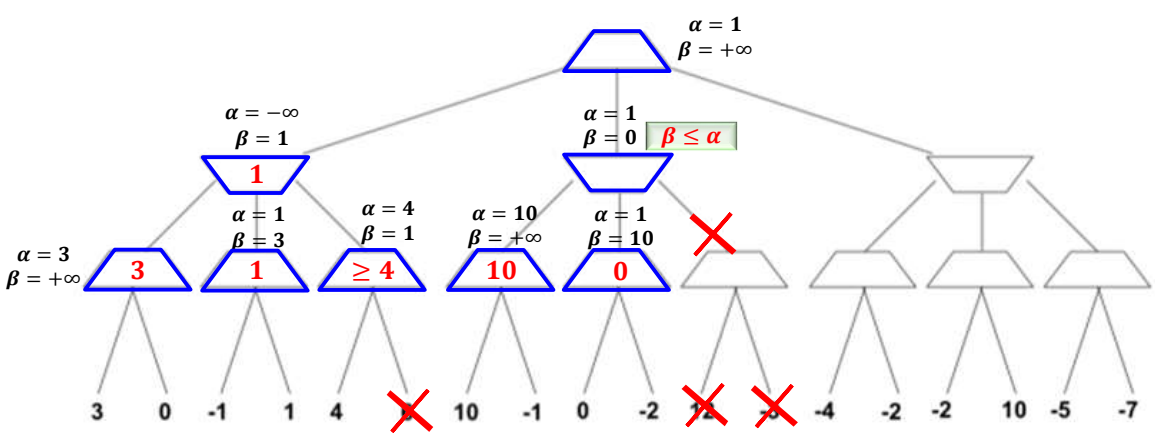


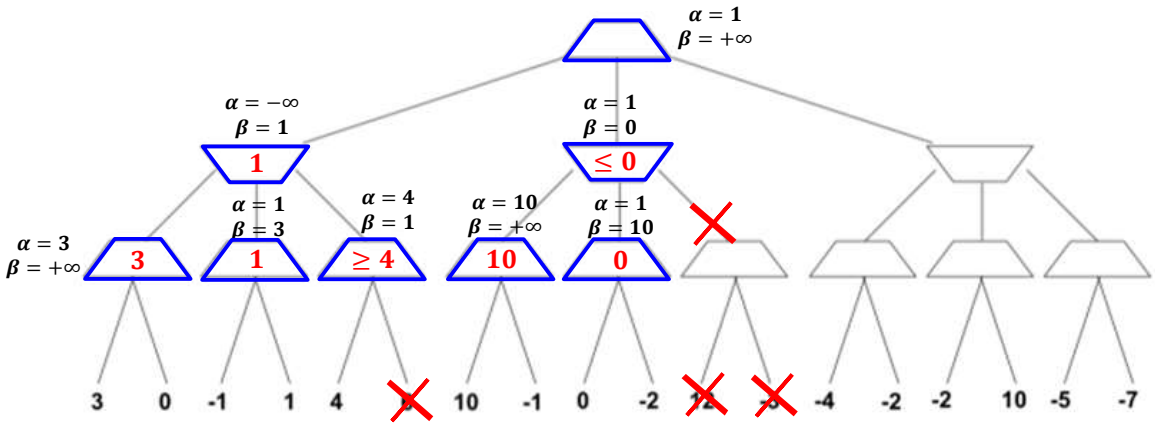



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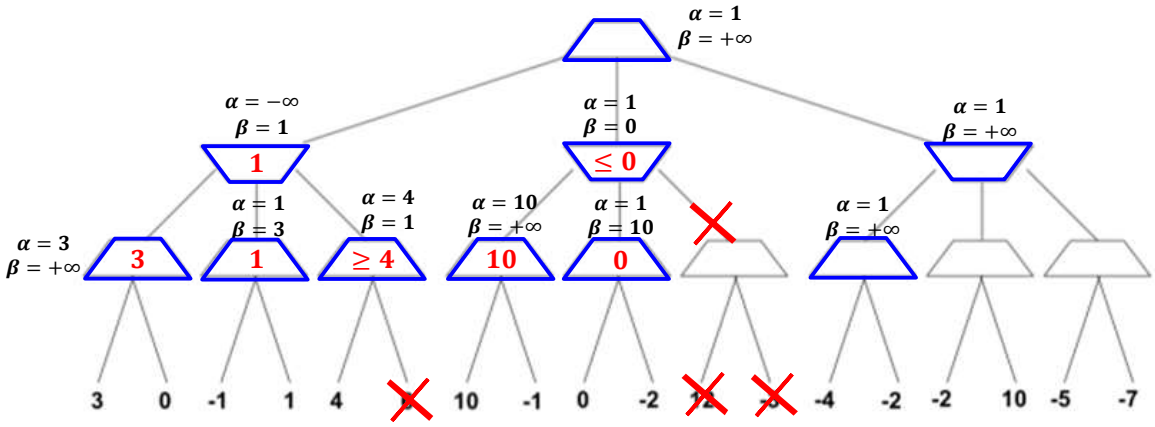



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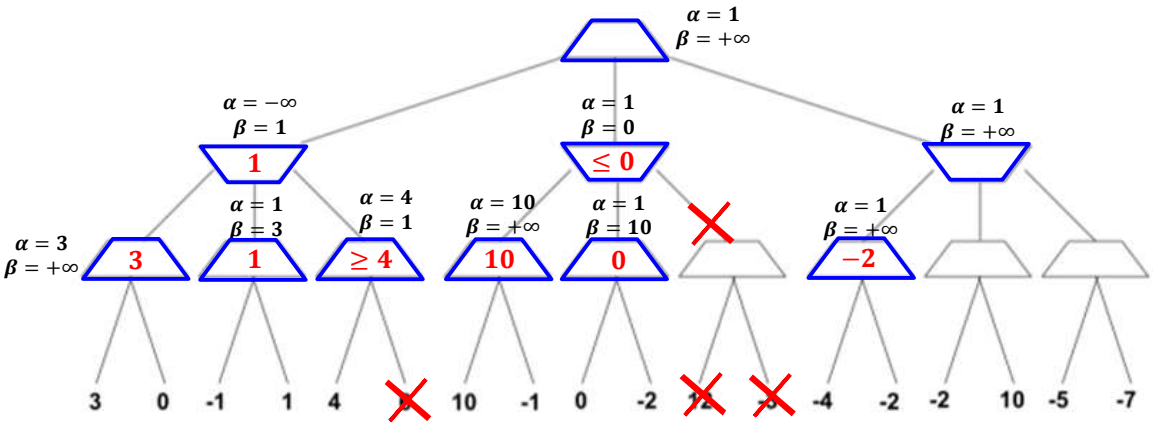





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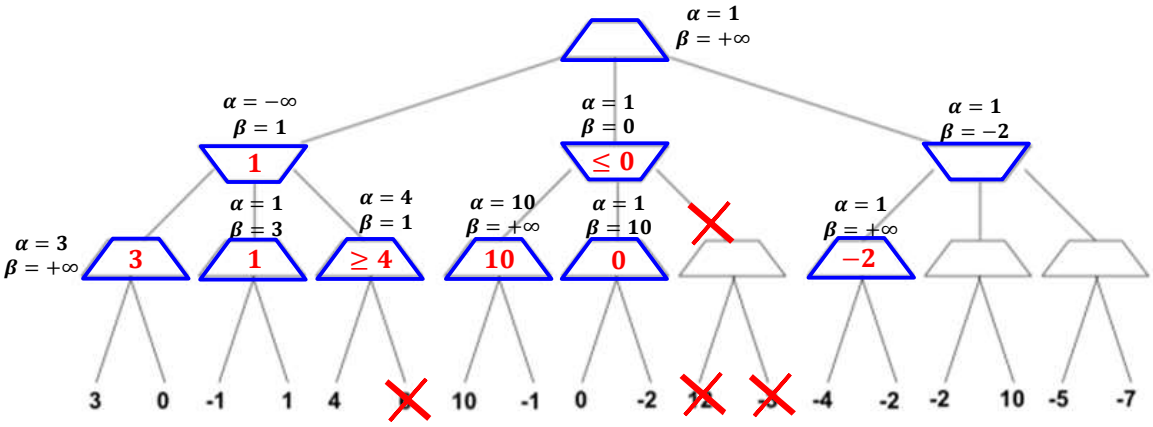


# Solution



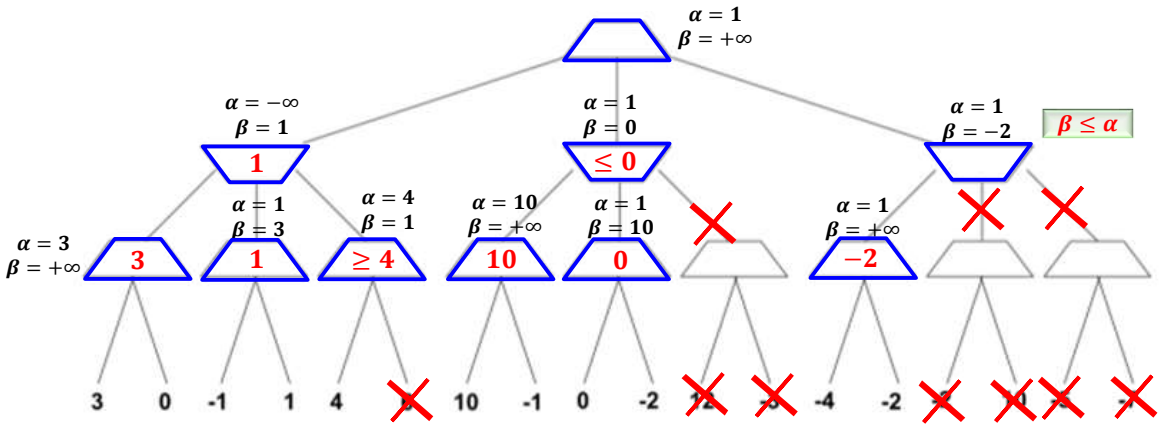



# Solution



# Solution

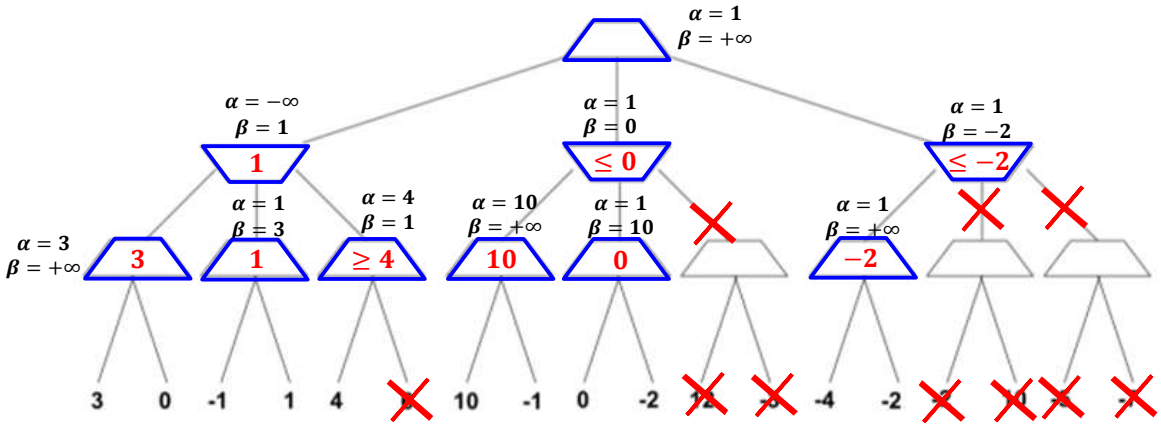
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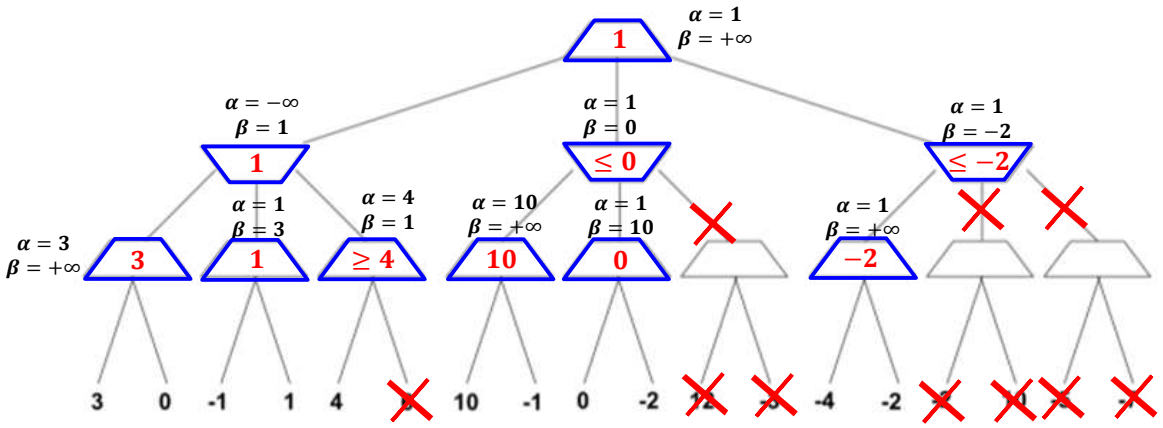



# Solution

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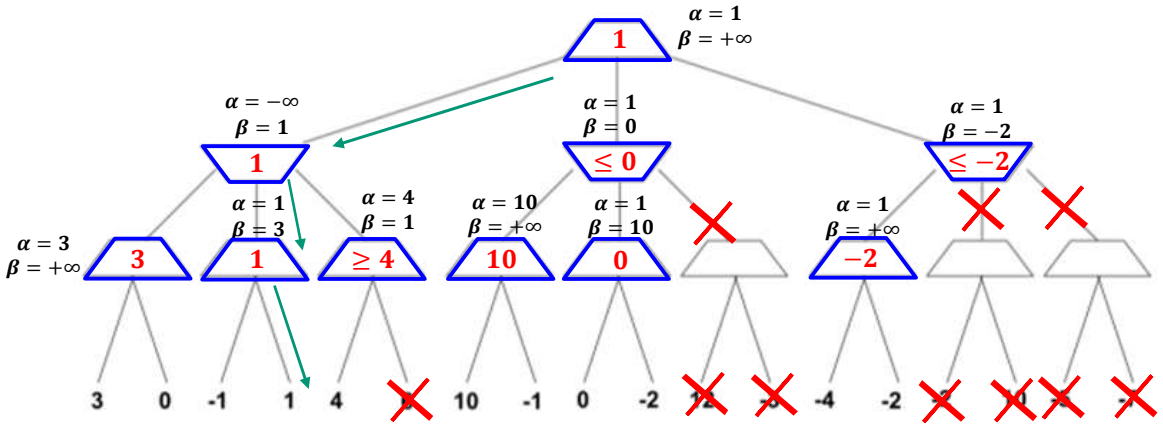


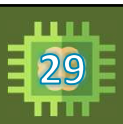
# Solution






# Solution






# Constraint Satisfaction Problems (CSPs)

Slides adapted from Prof. Tuomas Sandholm
Carnegie Mellon University, Computer Science Department



# Course Structure/Syllabus




| Syllabus | |
|---|---|
| **Module** | **Topics to be Covered** |
| **Module-1** | **INTRODUCTION TO AI:**<br>Artificial Intelligence Introduction, Brief History, Intelligent Agents, Types of agents<br>***Python Primer for AI*** |
| **Module-2** | **PROBLEM SOLVING BY SEARCH:**<br>Problem formulation, Concept of state space search<br>Introduction to Uninformed Search Techniques: *Breadth First Search, Depth First Search, Depth First Search with Iterative Deepening, Uniform Cost Search*<br>Introduction to Informed/Heuristic search techniques: *Greedy Best First Search, A* search, Hill Climbing search, Simulated Annealing search*<br>Introduction to GA, GA Operations: *Selection, Crossover, Mutation*<br>***Solving N-Queen Problem using GA***<br>Adversarial Search: *Minimax algorithm, Alpha-Beta Pruning*<br>***Building a bot to play tic-tac-toe/Building 8-puzzle solver*** |
| **Module-3** | **CONSTRAINT SATISFACTION PROBLEM (CSP):**<br>Introduction to CSP, Constraint Graph, Binary and Higher order CSP, Backtracking Search, MRV heuristic, Degree heuristic, Least Constraining-value heuristic, Forward Checking, Arc Consistency, Min-Conflicts Algorithm<br>***Solving map coloring problem, Solving puzzle*** |

To be discussed after the mid-semester exam

# Constraint satisfaction problems (CSPs)

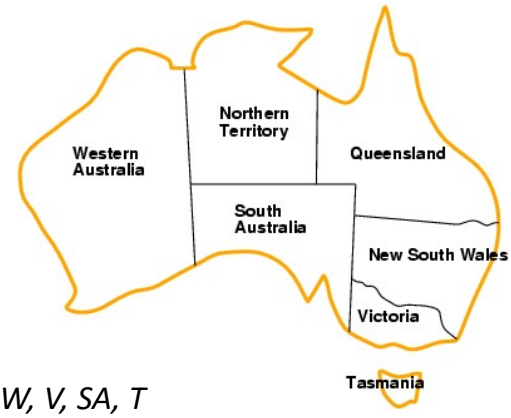



- Standard search problem:
  - **state** is a "black box" – any data structure that supports successor function and goal test
- CSP:
  - **state** is defined by variables $X_i$ with values from domain $D_i$
  - A set of constraints specifying allowable combinations of values for subsets of variables
- Allows useful general-purpose algorithms with **more power** than standard search algorithms



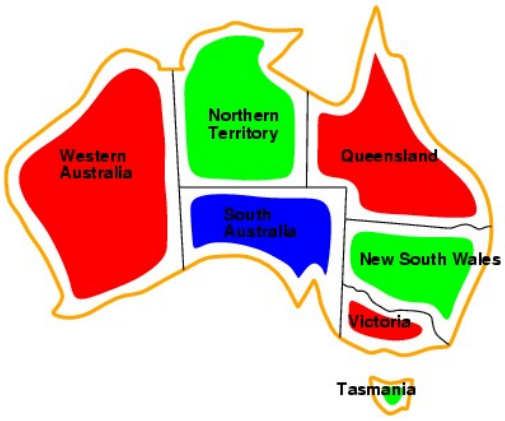
# Example: Map-Coloring




- Variables: *WA, NT, Q, NSW, V, SA, T*
- Domains: $D_i$ = {red, green, blue}
- Constraints: adjacent regions must have different colors
- e.g., WA ≠ NT, or (WA,NT) in {(red, green),(red, blue),(green, red),(green, blue), (blue, red),(blue, green)}

# Example: Map-Coloring




- Solutions are complete and consistent assignments
- e.g., WA = red, NT = green, Q = red, NSW = green, V = red, SA = blue, T = green



# Varieties of CSPs

- **Discrete variables**
  - finite domains:
    - $n$ variables, domain size $d \rightarrow O(d^n)$ complete assignments
    - e.g., Boolean CSPs, incl. Boolean satisfiability
  - infinite domains:
    - integers, strings, etc.
    - e.g., job scheduling, variables are start/end days for each job
- **Continuous variables**
  - e.g., start/end times for Hubble Space Telescope observations

# Varieties of constraints



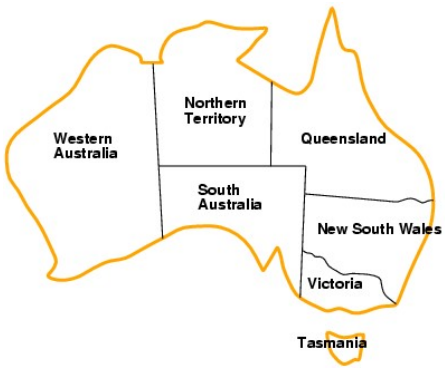

- **Unary** constraints involve a single variable,
  - e.g., SA ≠ green
- **Binary** constraints involve pairs of variables,
  - e.g., SA ≠ WA
- **Higher-order** constraints involve 3 or more variables,
  - e.g., cryptarithmetic column constraints

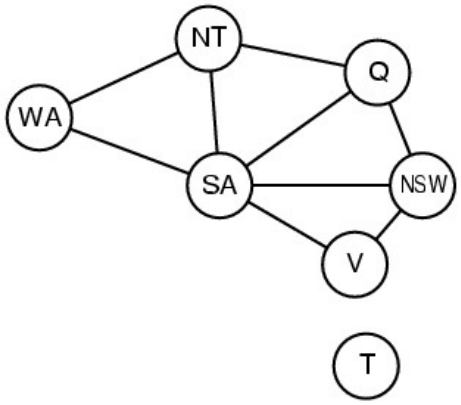


# Constraint Graph




- Binary CSP: each constraint relates two variables
- Constraint graph: nodes are variables, arcs are constraints



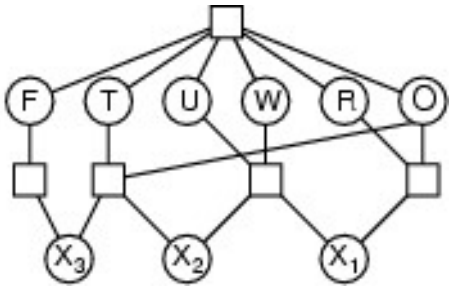



# Constraint Hypergraph




**Example: Cryptarithmetic Puzzle**

```
  T W O
+ T W O
-------
F O U R
```




- **Variables:** *F T U W R O* $X_1$ $X_2$ $X_3$
- **Domains:** {*0,1,2,3,4,5,6,7,8,9*}
- **Constraints:** *Alldiff (F,T,U,W,R,O)*

$O + O = R + 10 \cdot X_1$
$X_1 + W + W = U + 10 \cdot X_2$
$X_2 + T + T = O + 10 \cdot X_3$
$X_3 = F$, $T \neq 0$, $F \neq 0$



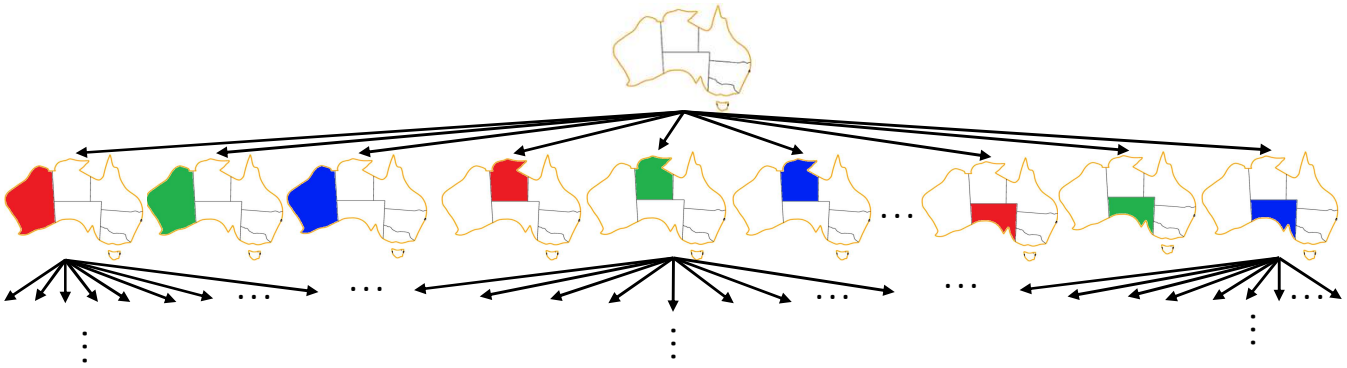
# Standard Search Formulation




- Standard search formulation of CSPs
- States defined by the values assigned so far (partial assignments)
  - **Initial state:** the empty assignment, {}
  - **Successor function:** assign a value to an unassigned variable
  - **Goal test:** the current assignment is complete and satisfies all constraints
- We'll start with the straightforward, naïve approach, then improve it

# Breadth First Search

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… All possible first variables
Check: Is there a solution?

**We generate a tree with $n!\, d^n$ leaves!**

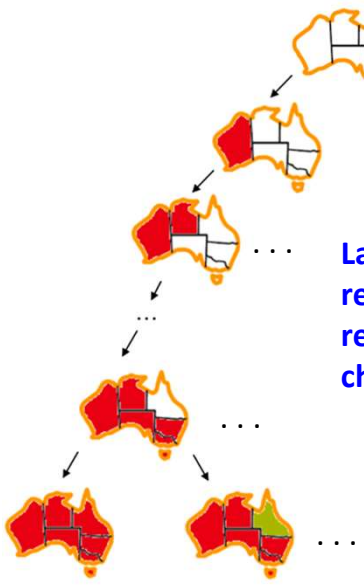






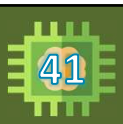

# Depth First Search

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**Late realization regarding requirement of change!**

**How CSP resolves this?**
**…. To be discussed next**




# Questions?